



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Symbiosis International (Deemed University)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Assignment No. 12

Subject:	DevOps Lab
TCODE:	TE7950
Name of Student	Adarsh Jha
PRN No.	23070122261
Branch	CSE, Batch (2023-27), L4
Academic Year & Semester	25 – 26, Sem VII
Date of Performance	23 rd July, 2026
Title of Assignment:	Kubernetes Pod & Deployment
Theory:	Kubernetes is an open-source container orchestration platform used to automate the deployment, scaling, and management of containerized applications. A Pod is the smallest deployable unit in Kubernetes and contains one or more containers that share the same network and storage resources. A Deployment manages Pods by maintaining the desired number of replicas, automatically replacing failed Pods, and supporting updates without downtime. Kubernetes simplifies application management by ensuring high availability, scalability, and efficient resource utilization across a cluster. Commands Used • Checked the Kubernetes client version: kubectl version --client • Verified the Kubernetes cluster information: kubectl cluster-info • Displayed the available cluster nodes: kubectl get nodes • Created the Kubernetes Deployment from the YAML manifest: kubectl apply -f web-deployment.yaml • Listed the running Pods: kubectl get pods • Deleted the Kubernetes Deployment and its Pods: kubectl delete -f web-deployment.yaml

Implementation:

Step 1: Enable Kubernetes in Docker Desktop

Enabled Kubernetes — Enabled Kubernetes in Docker Desktop and verified that the local Kubernetes cluster was successfully initialized.

Step 2: Verify kubectl

Verified Kubernetes Cluster — Verified that the local Kubernetes cluster was running successfully using kubectl cluster-info and confirmed that the control-plane node was in the Ready state using kubectl get nodes.

Step 3: Create the Kubernetes Deployment YAML

Created Kubernetes Deployment Manifest — Created the web-deployment.yaml file inside the k8s-manifests directory, defining a Kubernetes Deployment with two replicas of the Flask web application using the Docker image my-flask-app:v2.

Step 4: Apply the Deployment Manifest

Applied Deployment Manifest — Applied the Kubernetes Deployment manifest using kubectl apply, creating the Deployment resource in the local Kubernetes cluster.

Step 5: Verify the Pods

Verified Running Pods — Verified that the Kubernetes Deployment successfully created two running Pods, each hosting an instance of the Flask web application.

Step 6: Clean Up

Deleted Kubernetes Deployment — Removed the Kubernetes Deployment and the Pods it created using the kubectl delete command, cleaning up the local Kubernetes cluster.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\System32> kubectl version --client
Client Version: v1.36.3
Kustomize Version: v5.8.1
PS C:\WINDOWS\System32> choco install minikube
Chocolatey v2.5.1
Chocolatey detected you are not running from an elevated command shell
(cmd/powershell).

You may experience errors - many functions/packages
require admin rights. Only advanced users should run choco w/out an
elevated shell. When you open the command shell, you should ensure
that you do so with "Run as Administrator" selected. If you are
attempting to use Chocolatey in a non-administrator setting, you
must select a different location other than the default install
location. See
https://docs.chocolatey.org/en-us/choco/setup#non-administrative-install
for details.

Do you want to continue?([Y]es/[N]o): y

Installing the following packages:
minikube
By installing, you accept licenses for the packages.
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading Minikube 1.38.1... 100%
```

```
11 spec:
12   ports:
13     - port: 5000
14       targetPort: 5000
15   targetPort: 5000
16   nodePort: 30001
```

```
PS C:\Users\Adarsh Jha\Desktop\sem VII\DevOps-Lab-Assignments\23070122261_AdarshJha> cd Assignment-TW2-Advanced-DevOps
PS C:\Users\Adarsh Jha\Desktop\sem VII\DevOps-Lab-Assignments\23070122261_AdarshJha\Assignment-TW2-Advanced-DevOps> kubectl apply -f k8s/deployment.yaml
deployment.apps/flask-deployment created
PS C:\Users\Adarsh Jha\Desktop\sem VII\DevOps-Lab-Assignments\23070122261_AdarshJha\Assignment-TW2-Advanced-DevOps> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
flask-deployment-58669547d-q5qps    0/1     ContainerCreating    0          15s
flask-deployment-58669547d-xnf5z    0/1     ContainerCreating    0          15s
PS C:\Users\Adarsh Jha\Desktop\sem VII\DevOps-Lab-Assignments\23070122261_AdarshJha\Assignment-TW2-Advanced-DevOps> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
flask-deployment-58669547d-q5qps    0/1     ContainerCreating    0          31s
flask-deployment-58669547d-xnf5z    0/1     ContainerCreating    0          31s
PS C:\Users\Adarsh Jha\Desktop\sem VII\DevOps-Lab-Assignments\23070122261_AdarshJha\Assignment-TW2-Advanced-DevOps>
```

Subject Faculty Name: Prof. Arijit Dutta